

MAPIN 0.4MG/ML INJECTION (1ML AMP)

DESCRIPTION:

MAPIN 0.4MG/ML INJECTION (1ML AMP): Mapin Injection is a clear, colourless solution.

COMPOSITION:

Each ampoule contains Naloxone Hydrochloride 0.4mg/ml.

PHARMACODYNAMICS:

Naloxone hydrochloride is essentially a pure opiate antagonist; it has little or no agonistic activity. Naloxone is thought to act as a competitive antagonist at mu, kappa, and sigma opioid receptors in the central nervous system (CNS). It can prevent or reverse the effects of opioids including respiratory depression, sedation and hypotension. It can also reverse the respiratory depression caused by pentazocine.

PHARMACOKINETICS:

When naloxone is administered intravenously, the effects are usually apparent within 2 minutes; the onset of action is only slightly less rapid when it is administered intramuscularly or subcutaneously. The duration of action is dependent upon the dose and the route of administration but is usually in the region of 1 to 4 hours. Intramuscular injection produces a more prolonged effect than intravenous administration. Following parenteral administration, naloxone is rapidly distributed in the body. It is metabolised in the liver primarily by glucuronide conjugation and the metabolites are excreted in the urine. It is 50% protein bound. The elimination half-life in adults is 60-90 minutes and 180 minutes in neonates.

In the absence of narcotics or agonists effects of other narcotic antagonists, it exhibits essentially no pharmacologic activity.

Naloxone has not been shown to produce tolerance nor to cause physical or psychological dependence. In the presence of physical dependence on narcotics, naloxone will produce withdrawal symptoms.

INDICATION:

Naloxone is indicated for the complete or partial reversal of narcotic depression including respiratory depression induced by opioids including natural and synthetic narcotics, propoxyphene, methadone, and the narcotic-antagonist analgesics nalbuphine, pentazocine and butorphanol. Naloxone is also indicated for the diagnosis of suspected acute opioid overdose.

RECOMMENDED DOSAGE:

Naloxone hydrochloride may be administered intravenously, intramuscularly or subcutaneously. The most rapid onset of action is achieved by intravenous administration and it is recommended in emergency situations.

Since the duration of action of some narcotics may exceed that of naloxone, the patient should be kept under continued surveillance and repeated doses of naloxone should be administered, as necessary.

Intravenous Infusion

Naloxone may be diluted for intravenous infusion in normal saline or 5% dextrose solutions. The addition of 2mg of naloxone in 500ml of either solution provides a concentration of 0.004mg/ml. Infusion should be commenced as soon as practicable after preparation of the mixture in order to reduce microbiological hazards. Preparations not used within 24 hours should be discarded. The rate of administration should be titrated in accordance with the patient's responses.

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration whenever solution and container permit. Naloxone should not be mixed with preparations containing bisulphite, metabisulphite, long chain or high molecular weight anions, or any solution having an alkaline pH. No drug or chemical agent should be added to naloxone unless its effect on the chemical and physical stability of the solution has first been established.

Usage in Adults: Narcotic Overdose - Known or Suspected:

An initial dose of 0.4mg to 2mg of naloxone may be administered intravenously. If the desired degree of concentration and improvement in respiratory functions is not obtained, it may be repeated at 2 to 3 minutes intervals. If no response is observed after 10mg of naloxone have been administered, the diagnosis of narcotic induced or partial narcotic toxicity should be questioned. Intramuscularly or subcutaneous administration may be necessary if the intravenous route is not available.

Postoperative Narcotic depression

For the partial reversal of narcotic depression following the use of narcotics during surgery smaller doses of naloxone are usually sufficient. The dose of naloxone should be titrated according to the patient's response. Naloxone should be injected in increments of 0.1 to 0.2mg intravenously at two to three times intervals to the desired degree of reversal -i.e. adequate ventilation and alertness without significant pain or discomfort. Larger than necessary dosage of naloxone may result in significant reversal of analgesia and increase in blood pressure. Similarly, too rapid reversal may induce nausea, vomiting, sweating, or circulatory stress.

Repeat doses of naloxone may be required within one to two hours intervals depending upon the amount, type (i.e. short- or long-acting) and time interval since last administration of narcotic. Supplemental intramuscular doses have been shown to produce a longer lasting effect.

Usage in children: Narcotic Overdose - Known or Suspected:

The usual initial dose in children is 0.01mg/kg body weight given I.V. If this dose does not result in the desired degree of clinical improvement, a subsequent dose of 0.1mg/kg body weight may be administered. If an I.V. route of administration is not available, naloxone may be ministered I.M. or S.C. in divided doses. If necessary, naloxone can be diluted with sterile water for injection.

Postoperative Narcotic depression.

Follow the recommendations and cautions under "Adult Postoperative Depression". For the initial reversal of respiratory depression naloxone should be injected in increments of 0.005mg to 0.01mg intravenously at two to three minutes intervals to the desired degree of reversal.

Usage in Neonates: Narcotic-Induced Depression

The usual initial dose is 0.01mg/kg bodyweight administered i.v., i.m. or s.c. This dose may be repeated in accordance with adult administration guidelines for post-operative narcotic depression.

ROUTE OF ADMINISTRATION:

For i.m./s.c./i.v. injection.

CONTRACINDICATIONS:

Naloxone is contraindicated in patients known to be hypersensitive to it.

WARNINGS & PRECAUTIONS:

Naloxone should be administered cautiously to persons (including newborns of mothers) who are known or suspected to be physically dependent on opioids and who have received large doses of opioids; in such cases an abrupt and complete reversal of narcotic effects may precipitate an acute abstinence syndrome.

The patient who has satisfactorily responded to naloxone should be kept under continued surveillance and repeated doses of naloxone should be administered, as necessary, since the duration of action of some narcotics may exceed that of naloxone.

Following the use of opioids during surgery, excessive doses of naloxone hydrochloride should be avoided as they may cause excitement, increase blood pressure and reverse analgesia. A too rapid reversal of opioid effects may cause nausea, vomiting, sweating or tachycardia. Naloxone is not effective against respiratory depression due to non-opioid drugs.

In addition to naloxone, other resuscitative measures such as maintenance of a free airway artificial, ventilation, cardiac massage and vasopressor agents should be available and employed when necessary to counteract acute narcotic poisoning.

Several instances of hypotension, hypertension, ventricular tachycardia and fibrillation and pulmonary oedema have been reported. These have occurred in postoperative patients, most of who had pre-existing cardiovascular disorders or received other drugs which may have similar adverse cardiovascular effects.

Although a direct cause-and-effect relationship has not been established naloxone should be used with caution in patients with pre-existing cardiac disease or patients who have received potentially cardiotoxic drugs.

INTERACTION WITH OTHER MEDICAMENTS:

None Available

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Naloxone should not be used in pregnant women only when clearly needed.

Use in Lactation: It is not known whether Naloxone is excreted in human milk. Because many drugs are excreted in human milk, caution should be exercised when naloxone is administered to a nursing woman

SIDE EFFECTS:

Abrupt withdrawal of narcotic depression may result in nausea, vomiting, sweating, tachycardia, increased blood pressure and tremulousness. In postoperative patients, larger than necessary dosages of naloxone may result in significant reversal of analgesia and in excitement. Hypotension, hypertension, ventricular tachycardia and fibrillation, and pulmonary oedema have been associated with the use of naloxone postoperatively. (See Precautions, and Usage in Adults, Postoperative Narcotic Depression). Seizures have been reported to occur infrequently after the administration of naloxone; however, a causal relationship has not been established.

SYMPTOMS AND TREATMETN OF OVERDOSE:

There is no clinical experience with naloxone overdosage in humans.

INCOMPATIBILITIES:

Naloxone should not be mixed with preparations containing bisulphite, metabisulphite, long-chain or high molecular weight anions or any solution having an alkaline pH. No drug or chemical agent should be added to naloxone unless its effect on the chemical and physical stability of the solution has first been established.

STORAGE CONDITIONS:

Store below 30°C. Protect from light. Keep out of reach of children. Jauhkan daripada kanak-kanak.

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

Packs of 10's, 30's, 50's, and 100's in 1ml amber ampoules

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.

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